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(PROPOSED)
MILITARY SPECIFICATION

DRIVE ASSEMBLY, CONSTANT SPEED, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the general requirements for design, performance and test of aircraft constant speed drive assemblies which are mounted on engine accessory drive pads or remotely located gear mechanisms. These constant speed drives could transmit power by mechanical or hydraulic components. A model specification shall be prepared by the airframe contractor or drive manufacturer to more adequately define performance and quality assurance requirements in conjunction with this specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-S-365	Silver Plating, Electrodeposited; General Requirements for
QQ-Z-325	Zinc Coating, Electrodeposited; Requirements for
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof-for Packaging and Sealing

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Engineering and Associated Lists
MIL-C-5015	Connectors, Electric, AN Type
MIL-C-5501	Caps and Plug, Protective, Dust and Moisture Seal

FSC 1650

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MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-D-6054	Drums, Metal Shipping and Storage
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-9282	Container, Shipping, Metal Reusable
MIL-T-10727	Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Non-Ferrous Metals
MIL-F-22191	Films, Transparent, Flexible, Heat Sealable, for Packaging Applications
MIL-B-25305	Box, Metal, or Nonmetal, Shipping
MIL-E-25366	Electric and Electronic Equipment and Systems, Guided Missiles, Installation of, General Specification for
MIL-E-25499	Electrical Systems, Aircraft, Design and Installation of, General Specification for
MIL-C-26074	Coating, Nickel-Phosphorous, Electroless Nickel Requirements for
MIL-C-26500	Connectors, General Purpose, Electrical, Miniature, Circular, Environmental Resisting, 200°C, Ambient Temperature.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

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3. REQUIREMENTS

3.1 Precedence. The requirements of MIL-STD-704, MIL-E-25499, and MIL-E-25366, as applicable, apply as requirements of this specification with the exceptions and additions specified herein. In case of conflict between the requirements of this specification and the model specification, the requirements of the model specification shall take precedence.

3.2 Functional requirements

3.2.1 Lubrication system. A self-contained lubrication system shall be included. The oil reservoir, oil filter and cooler (if required) shall be considered composite parts of the CSD lubrication system.

3.2.1.1 Lubricating oil. Lubricating oil for each drive model shall be specified in the model specification. The oil selected for this drive shall not require changing more often than every 200 hours. No changes in lubricants shall be required for starting and operation throughout the complete ground and air temperature range.

3.2.1.2 Lubrication system design. Design provisions in the use of the lubricant shall be stated herein. Where lubricant is used, other than for cooling working surfaces (bearings, gear teeth, splines, rubbing seals, etc.), the temperature of surfaces to which lubricant and lubricant vapors are exposed shall be stated and the location or usage identified.

3.2.1.3 Performance

3.2.1.3.1 Oil pressure and temperature. The operating oil pressure and temperature limits and maximum transient oil temperatures shall be specified in the model specification. The minimum and maximum oil pressures during starting and initial operation at cold weather conditions prior to significant oil temperature increase shall be specified in the moel specification. Minimum and maximum oil pressures during starting and initial operation shall not persist for more than 2.5 minutes.

3.2.1.3.2 Oil supply. A dip stick or other oil level system device shall be furnished. Dimensions and details of these provisions shall be shown on the installation drawing.

3.2.1.3.3 Oil comsumption. The oil consumption shall not exceed the following: (Contractor shall insert consumption rate consistent with tank capacity, lubrication system and aircraft endurance missions(s).)

____cc/hr average during preproduction tests.

____cc/hr average during quality conformance tests.

____cc/hr average during quality re-evaluation periodic tests.

____cc/hr average during service use.

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If the average oil consumption rate during the preproduction tests, the quality conformance tests or the quality re-evaluation periodic tests is less than one third of the specified value, the specification oil consumption rate shall be adjusted to a value no greater than three times the preproduction tests, quality conformance test and the quality conformance test and the quality re-evaluation periodic tests. The average oil consumption is the oil consumption in cubic centimeters per hour during all conditions of operation, transients, shutdowns, etc.

3.2.1.4 Scavenging system. The scavenging system shall adequately scavenge the constant speed drive under ground operating conditions and under all flight conditions. Back pressure on the scavenging system shall be defined in the model specification.

3.2.1.5 Oil drains. Provisions shall be made for draining the constant speed drive of oil while the CSD is in a horizontal position, 15 degrees nose up, and 20 degrees nose down. When so specified in the model specification, provisions shall be made for draining the CSD of oil while the CSD is in the vertical position. Magnetic chip detectors, of a design providing a warning signal output, shall be installed in all oil drain ports or in locations where the warning signal will be most significant. The electrical characteristics of these chip detectors shall be described in the model specification. Oil drain provisions to determine wear condition indication through oil spectrographic analysis is required. The specific time interval for taking oil samples for spectrographic analysis shall be specified in the model specification. The threshold limits of specific metallic elements shall be specified in the model specification showing normal and abnormal rate of wear limits.

3.2.1.6 Internal oil leakage. The design of the lubricating system shall be such that oil leakage within the constant speed drive shall not cause oil discharge from the CSD upon subsequent starting after shutdown, adversely affect oil supply determination, or contribute to objectionable deposit buildup or lubricant contamination over specified time between overhaul (TBO).

3.2.2 Rating. The horsepower rating of the drive assembly at the designed output speed shall be specified in the model specification. The drive assembly shall be capable of delivering 150 percent of rated horsepower for 2 minutes.

3.2.3 Protective devices. The drive assembly shall have internal protection that will prevent damage to itself from malfunction of the aircraft engine and the aircraft electrical system. Failure of the drive governor or transmission shall not render the aircraft engine inoperative. The protective devices shall be specified in the model specification.

3.2.3.1 Overspeed protection. The drive shall be equipped with an overspeed protective device, which shall automatically operate in the event of a malfunction of the normal governor system which results in the generator drive unit operating at output speeds in excess of the normally governed values. The overspeed protective device shall operate by the time that the drive output speed reaches a value corresponding to 500 cycles per second as a maximum. The overspeed protective device shall also

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be such that the maximum sustained operating output speed of the drive necessary to effect operation of the protective device will not exceed a speed corresponding to 464 cycles per second in the event of such failure. The protective device shall also limit the speed of the generator to a speed corresponding to 500 cycles per second as a maximum by placing the drive in full underdrive or by other suitable means, upon actuation.

3.2.3.2 Underspeed protection. (Parallel operation applications only.) The drive assembly shall be equipped with an underspeed protective device, separate from the control governor. This device shall disconnect the generator from the load bus similar to the overspeed switch. The drive speed ratio need not be changed as a result of this underspeed condition.

3.2.4 Drive disconnect. An input shaft shall be incorporated in the drive assembly which could be disengaged from the engine coupling in the event malfunctions occur in the drive system or the associated power generating system. The drive disconnect shall be subjected to the drive disconnect test specified in 4.3.2.1.15. Re-engagement of the disconnect shall be only accomplished on the ground when the engine or remotely located gear box mechanism is not rotating. This should be accomplished without disassembly or dismounting of the constant speed drive in the case of a manually operated device.

3.2.4.1 Warning devices. The drive assembly shall contain one or more warning devices which can switch on a remote warning light to signal need for disconnect. An overheat-indicating switch is mandatory. Other warning devices are optional.

3.2.5 Motoring. (Parallel operation applications only.) A means shall be incorporated into the drive assembly to prevent the generator from motoring the drive.

3.2.6 Oil leakage. There shall be no external seepage of oil at a rate sufficient to form a drip under any of the drive operational conditions with the exception that the shaft seal leakage shall not exceed 3cc per hour.

3.2.7 Shear section. A shear section shall be included in the input shaft of the constant speed drive. After a shaft shear, there shall be no damage to the engine or remotely located gear box mechanism and shearing shall not contaminate the oil systems. A shaft shear caused by a generator malfunction shall not cause damage to the constant speed drive except for the sheared shaft.

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3.3 Installation requirements

3.3.1 Overboard drains. All external shaft seal leakage shall be vented overboard.

3.3.2 Oil filters. A suitable oil filter shall be provided in the drive charge system. The micron rating and details of this filter shall be specified in the model specification for this drive. A differential pressure indicator, which is temperature compensated, shall be incorporated in this filter assembly. Also, under operating conditions described in this specification, the filter shall not require servicing more than once every 200 hours.

3.3.3 Anti-drain. If a separate reservoir is used, the drive assembly shall incorporate on the pressure side of the charge pumps a means to prevent oil in the reservoir from draining into the drive when the drive is not operating.

3.3.4 Mounting. The drive mounting provisions shall conform dimensionally to one of the accessory flange or drive standards developed by the Society of Automotive Engineers Technical Committee AE-1 and shall be of the quick attach-detach type. Complete dimensions and details shall be shown on installation drawings to be furnished with the model specification.

3.3.5 Electrical connectors. The drive assembly electrical connectors shall be in accordance with MIL-C-5015 or MIL-C-26500.

3.3.6 Weight. The weight of the drive assembly shall be specified in the model specification.

3.3.7 Rotation. The direction of rotation of the drive shall be clearly and permanently marked on an exposed surface of the drive housing.

3.3.8 Dimensions. Dimensions pertinent to the installation of constant speed drives in aircraft shall be specified on the manufacturer's installation drawing.

3.3.9 Ports. All ports shall be identified by clear and permanent markings.

3.3.10 Heat rejection. The over-all efficiency of the drive should not be less than 75 percent at rated operating conditions. The CSD manufacturer shall provide the drive parameters necessary to design the heat exchanger. These parameters shall be included in the model specification.

3.3.11 Parallel operation. When two or more a-c generators driven by constant speed drive assemblies are opeated in parallel, tests should be conducted in accordance with the requirements of 4.3.2.1.6, 4.3.2.1.7 and 4.3.2.1.9.

3.4 Environmental requirements. In general, constant speed drives shall be designed to operate without limitations as to time and without impairment of function or change in adjustment, under environmental conditions as specified in the following subparagraphs:

3.4.1 Altitude. The drive performance shall not be affected by change in altitude up to the maximum as specified in the model specification for this drive.

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3.4.2 Temperature. The operating temperature range of the drive and lubricating oil shall be specified in the model specification.

3.4.3 Vibration. The drive assembly shall perform satisfactorily under the range of vibration to which it will be excited when mounted on the installed or stand-mounted engine(s) with the generator(s) to be used in conjunction with this drive.

3.4.3.1 As part of the design integrity test, the drive shall be subjected to the vibration test specified in 4.3.2.1.13.6.

3.4.4 Acceleration. The constant speed drive shall be designed to withstand sustained accelerations in any direction. The magnitude of these accelerations shall be specified in the model specification.

3.4.5 Atmospheric conditions. The constant speed drive shall be designed to withstand continuous exposure, in the configuration as installed in aircraft, either operating or nonoperating, to salt spray as encountered in marine or coastal areas and to sand and dust as encountered in desert areas.

3.5 Detail requirements

3.5.1 Material. Materials and processes used in the manufacture of these constant speed drives shall be of high quality, suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to the drive manufacturer's material specifications may be used provided the specifications are acceptable to the procuring activity and contain provision for adequate tests. The use of the drive manufacturer's specification will not constitute waiver of other applicable specifications.

3.5.1.1 Metals. All metals shall be compatible with the intended temperature, functional, service and storage conditions to which the drive components will be exposed. The metals which are not protected from corrosion by the lubricating oil shall be of the corrosion-resisting type, or shall be suitably treated to resist corrosion in accordance with 3.5.1.2.

3.5.1.2 Corrosion protection. Metals which do not possess adequate corrosion resisting characteristics shall be suitably protected in accordance with the following subparagraphs to resist corrosion as dissimilar metal combinations, moisture, salt spray and high temperature deterioration as applicable.

3.5.1.2.1 Ferrous and copper alloys. Ferrous alloys requiring corrosion preventive treatment, and all copper alloys, except for parts having bearing surfaces, shall have a suitable electrodeposited metallic coating selected from table I. Tin, cadmium, and zinc plating shall not be used for internal parts or on internal surfaces in contact with oil or exposed to its vapors, nor where subject to abrasion. Class and type are at the option of the manufacturer.

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TABLE I. Metallic Coatings

Coating	Specification
Cadmium plating	QQ-P-416
Zinc plating	QQ-Z-325
Chromium plating	QQ-C-320
Nickel plating	QQ-N-290
Silver plating	QQ-S-365
Tin plating	MIL-T-10727
Electroless nickel	MIL-N-26074

Other metallic coatings that have been approved by the procuring activity, such as electrodeposited 85 percent tin, 15 percent cadmium alloy, may be used. Unless otherwise specified by the procuring activity, all aluminum alloys shall be anodized in accordance with MIL-A-8625, except that in the absence of abrasive conditions they may be coated with chemical film in accordance with MIL-C-5541. The exceptions noted shall be subject to the approval of the procuring activity.

3.5.1.3 Castings. Castings shall be of high quality, clean, sound, and free from cracks, blow holes, and excessive porosity and other defects. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnation, welding, or other methods acceptable to the procuring activity. Inspection and repair of castings shall be governed by quality control techniques and subject to the approval of the procuring activity.

3.5.2 Nameplate. A nameplate containing the following information legibly filled in shall be securely attached to the drive. The information marked in the spaces provided shall be in accordance with MIL-STD-130.

Aircraft Constant Speed Drive

Model specification No._____

Mfrs. part No._____(or identification)

Mfrs. name or trademark_____

Serial number_____

Lubricating oil_____

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Rating:

Horsepower (or kva)_____

Output Speed_____rpm

3.5.2.1 Any nameplate data required in addition to the above shall be specified in the model specification.

3.5.3 Reliability. Unless otherwise specified in the model specification the CSD assemblies shall have a specified mean time between failures of 3000 hours, and a minimum acceptable mean time between failures of 1000 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. For the purpose of demonstrating compliance of the drives with this specification and the model specification, the following test program shall be conducted:

a.	Preproduction tests	(4.3)
b.	Quality conformance tests	(4.4)
c.	Reliability tests	(4.5)

4.2.1 Preproduction tests. Preproduction tests, for the purpose of demonstrating conformance of the drive design to the requirements of this specification and with the model specification, to insure their suitability for use in specific installations, and also to verify the manufacturer's ability to produce the item, shall consist of the design integrity tests specified in 4.3.2.

4.2.2 Quality conformance tests. Quality conformance tests for acceptance shall consist of a routine visual and dimensional examination and test program to determine conformance of the drives of a given model to the dimensional, workmanship, and performance requirements of this specification and the pertinent model specification. This program is specified in 4.4.

4.2.3 Reliability tests. The reliability tests shall consist of the tests specified in 4.5.1.2.

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4.3 Preproduction tests

4.3.1 Design approval procedure

4.3.1.1 Model specification. The airframe contractor or drive manufacturer shall prepare a model specification for each drive for which design approval is desired. This specification shall specify conformance with the requirements of this specification, including satisfactory completion of the design approval tests specified herein. In addition, it shall include any supplemental requirements and the detail test methods necessary to insure satisfactory operation and service life in the particular installation involved. This model specification shall be submitted to the applicable procuring activity for release.

4.3.1.2 Similarity. In cases where the drive for which design approval is desired incorporates the same working parts as a drive which has previously received design approval, all or a portion of the design approval tests may be waived, and a report substantiated by drawings showing similarity to the approved article submitted in lieu of actual tests.

4.3.1.3 Design approval test report. A report of the tests conducted and the results of the tests shall be compiled and submitted to the procuring activity. This report shall include full evaluation of the extent of compliance of the tested drives to the specified requirements. The report shall include full descriptions of the manner in which the tests were performed including instrumentation description, schematic diagrams, and photographs as may be appropriate. The report shall include complete copy of test data in tabular form. Test circuits shall be described in complete detail for each test. A complete set of detail and assembly drawings of the drive conforming to the requirements of MIL-D-1000 shall accompany the report.

4.3.2 Design integrity tests. The design integrity tests shall consist of the following tests, conducted in the order listed on one sample drive:

a.	Quality conformance tests	(4.3.2.1.1)
b.	Examination of product	(4.3.2.1.2)
c.	No load	(4.3.2.1.3)
d.	Rated load	(4.3.2.1.4)
e.	Speed regulation	(4.3.2.1.5)
f.	Parallel operation	(4.3.2.1.6)
g.	Load division	(4.3.2.1.7)
h.	Governor overspeed	(4.3.2.1.8)

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i. Underspeed switch	(4.3.2.1.9)
j. Low temperature starts	(4.3.2.1.10)
k. Heat rejection	(4.3.2.1.11)
l. Endurance	(4.3.2.1.12)
m. Environmental	(4.3.2.1.13)
n. Radio noise	(4.3.2.1.14)
o. Drive disconnect	(4.3.2.1.15)
p. Shear section	(4.3.2.1.16)

Note: To expedite completion of the design integrity tests, upon approval of the constant speed drive by the procuring activity, more than one sample may be used. If this option is selected, the quality conformance tests (4.4) must be conducted on each test sample.

4.3.2.1 Design integrity test methods. Unless otherwise specified, required test operating conditions shall be maintained within the following limits:

Pressure:	±2 percent
Temperature:	-65° to 140°F ±5°F
	140° to 240°F ±10°F
	240° to 400°F ±15°F
Flow:	±2 percent
Electrical:	±2 percent
Speed:	±5 rpm
Power:	±2 percent

4.3.2.1.1 Quality conformance tests. The quality conformance tests which are included in the design approval test program shall be run exactly as specified in 4.4.

4.3.2.1.2 Examination of product. The drive assembly shall be inspected to determine compliance with the requirements specified herein with respect to materials, dimensions, workmanship and marking.

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4.3.2.1.3 No load. The drive assembly output frequency shall remain at design frequency ±4 cps average value with not more than ±10 cps instantaneous deviation, while the input speed is accelerated from minimum drive design speed to 100 percent and back to minimum drive design speed. The rate of acceleration and deceleration should be approximately 50 rpm per second. This test shall be conducted twice. The no-load test shall be repeated except that the drive output frequency shall remain at design frequency ±20 cps and the rate of acceleration shall be 900 rpm per second minimum, with an input speed range of 75 percent to 100 percent of drive design speed.

4.3.2.1.4 Rated load. Rated load (horsepower) shall be imposed on the drive assembly. The input speed shall be accelerated from minimum drive design speed to 100 percent and back to minimum drive design speed at a rate of 400 to 600 rpm per second. The output frequency shall remain at design frequency ±20 cps. This test shall be repeated with 50 percent rated load (horsepower). The output frequency shall remain at design frequency ±20 cps.

4.3.2.1.5 Speed regulation. The input speed shall be maintained at minimum drive design speed. Loads of 25, 50, 75 and 100 percent of drive rated horsepower shall be applied in steps and the output frequency shall be recorded. The steady state output frequency shall remain at 100 percent drive design frequency ±4 cps (average value measured during a 1-minute period). The test shall be repeated at an input speed equivalent to 100 percent of drive design speed. The steady state output frequency shall remain at drive design frequency ±4 cps (average value measured during a 1-minute period).

4.3.2.1.6 Parallel operation (If applicable). Two a-c generators driven by constant speed drive assemblies shall be operated in parallel. The input speed to both assemblies shall be maintained at 95 percent of drive design speed. Load shall be applied to the drive assembly in 8 equal step increments up to rated drive horsepower (total for both drives). Division of load between the drive assemblies shall be equally maintained within 40 percent of the horsepower step input.

4.3.2.1.7 Load division (If applicable). Two generators powered by drive assemblies shall be connected in parallel with a total load of not less than 30 percent of the total drive rated horsepower (both drives). The input speed to one drive assembly shall be adjusted to approximately 95 percent drive design speed and the other to minimum drive design speed. The input speed to the latter shall then be accelerated to 100 percent drive design speed and decelerated back to minimum drive design speed three times at a rate of 300 to 500 rpm per second. This test shall be repeated with an acceleration rate of 900 to 1200 rpm per second, but with an input speed range of 75 to 100 percent of drive design speed. Throughout this test, the total load division shall be maintained within 20 percent of the drive rated horsepower.

4.3.2.1.8 Governor overspeed. The governor drive mechanism shall be disabled by suitable means while the drive is operating at maximum speed. The output speed shall immediately drop to 370 cps (maximum) and the underspeed devices (if so equipped) shall trip.

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4.3.2.1.9 Underspeed switch. The drive assembly output speed shall be increased from 0 to drive design speed. The underspeed switch contacts shall open at 96 percent of drive design speed maximum. The output speed shall then be decreased from drive design speed to zero. The switch contacts shall remain open to 96 percent of drive design speed and shall close before 90 percent of drive design speed minimum.

4.3.2.1.10 Low temperature starts. The drive assembly shall be coupled to a calibrated a-c generator and operated at minimum input drive design speed. The output frequency shall be at drive design frequency ±4 cps average value with no load on the generator. The drive assembly shall then be placed in a test chamber which will be cooled to and stabilized at a temperature as defined in paragraph 3.2.1.3.1. The oil used for the low temperature tests will be cooled to maintain the maximum oil viscosity plus 0, minus 2 percent of the maximum allowable viscosity specified in the oil specification. The drive assembly shall then be started and run at an input speed equal to the minimum drive design speed and no load. The output frequency shall be drive design frequency ±20 cps average value. At the end of 5 minutes operation, the output frequency shall be drive design frequency ±4 cps. The test shall be repeated with the drive started at minimum input speed and rated horsepower load. The output frequency shall be maintained at drive design frequency ±20 cps average value. After 5 minutes of operation, the output frequency shall be drive design frequency ±4 cps.

4.3.2.1.11 Heat rejection. Thermocouples shall be placed in the drive manifold inlet, discharge, and case scavenge openings in the adapter plate immediately adjacent to the drive. The drive shall then be completely enclosed with a coating of not less than 2 inches of 85 percent magnesia or equivalent insulation. Runs shall then be made at input speeds of 50, 75 and 100 percent drive design speed at rated horsepower load at each speed. The oil inlet temperature to the drive shall be maintained at normal operating temperature ±10°F during each run and the temperature differential and flow between the inlet, discharge, and case drain ports shall be measured to obtain the heat rejection in BTU's per minute. The efficiency of the drive shall be computed by means of the heat rejection data. The unit shall be operated for 5 minutes before each reading to insure stabilized flows and temperatures. As an alternate the efficiency can be measured and computed by means of the delta horsepower method.

4.3.2.1.12 Endurance. The drive assembly shall be operated for a minimum of 1000 hours at load and input speed cycles as specified in the following table.

TABLE II. Load and Speed Cycle for Drive Endurance Test

Time (hours)	Horsepower Rated (percent)	Input Speed Drive design
1.5	25	Maximum
3	50	Minimum
1	75	Minimum
3	100	Maximum
.5	105	Maximum
.5	0	Minimum

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The above test shall be conducted with ambient oil temperatures equal to the maximum rated lubricating oil temperature shown in the model specification. All acceleration and declerations shall be made at a rate of 400 ±50 rpm per second. If the load and speed schedule specified in Table II is not representative of the aircraft mission profile, the model specification shall specify an applicable load and speed schedule.

4.3.2.1.12.1 No frequency adjustment shall be made throughout the test. Average steady state frequency shall not drift more than ±4 cps. Incidental speed change necessary to maintain fixed frequency shall not exceed 50 rpm per second and electrical load variations shall not exceed ±5 percent due to voltage regulator, load resistance changes, et cetera.

4.3.2.1.12.2 Leakage from the drive assembly shall be recorded. Total external leakage from the drive assembly output shaft shall not exceed 3 cc per hour. Leakage from the transmission case shall not exceed 4 drops per hour.

4.3.2.1.12.3 During the endurance test,the drive assembly shall be subjected to a load equal to 150 percent of rated horsepower for a duration of 2 minutes once every 150 hours of operation (6 times total for this test). The input speed shall be 50 percent of drive design speed for this portion of the test.

4.3.2.1.12.4 During the endurance test, the drive assembly shall be subjected to a load equal to 200 percent of rated horsepower for a duration of 5 seconds. The input speed shall be 50 percent of drive design speed for this portion of the test.

4.3.2.1.12.5 Teardown inspection. After completion of the endurance test, the drive shall be disassembled and all parts visually inspected. The general condition of the parts shall be recorded.

4.3.2.1.13 Environmental. The drive assembly shall be subjected to the following tests. The drive assembly need not be operated after each environmental test individually, but shall pass the quality conformance tests specified herein after having been subjected to all the environmental tests. The environmental tests shall not cause adverse effects to the drive assembly that will cause malfunction and unnecessary maintenance.

4.3.2.1.13.1 Altitude. The drive assembly shall operate satisfactorily for 10 hours under simulated maximum altitude conditions. The maximum altitude shall be specified in the model specification. Oil leakage from the drive assembly shall not exceed the values specified in paragraph 3.2.6.

4.3.2.1.13.2 Humidity. The drive assembly shall be subjected to the humidity test specified in MIL-STD-810, procedure I.

4.3.2.1.13.3 Salt fog. The drive assembly shall be subjected to a salt fog test in accordance with MIL-STD-810, procedure I for a period of 50 hours. The magnesium or aluminum parts of the drive assembly shall not pit or shown any signs of transformation to a powder-like substance to. the extent that it will adversely affect

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operation of the drive. Corrosion shall not develop to the point of flaking or pitting that will in any way interfere or cause difficulty with the electrical conduction.

4.3.2.1.13.4 Fungus. The drive assembly shall be subjected to a fungus test in accordance with MIL-STD-810. Fungus growth shall not be such as to interfere with normal operation of insulation, materials, and contacts.

4.3.2.1.13.5 Sand and dust. The drive assembly shall be subjected to sand and dust in accordance with MIL-STD-810, procedure I.

4.3.2.1.13.6 Vibration. The vibration test shall be conducted in accordance with MIL-STD-810, procedure to be specified in the model specification, with the exception that the -65°F portion of this test should be accomplished at room temperature. The type and location of the vibration pickup(s) shall be specified on the installation drawing.

4.3.2.1.13.7 Explosion proof. Electrical components of the drive assembly shall be subjected to explosion-proof tests as specified in MIL-STD-810, procedure I. Any occurrence of an explosion shall be cause for rejection.

4.3.2.1.14 Radio noise. The generator drive and associated components shall be subjected to radio noise tests specified in the applicable paragraphs of MIL-STD-826. These tests shall be conducted at simulated transmission loads from 0 to rated load. The radio interference shall be measured during such conditions as starting, overspeed (including operation of overspeed device and eventual shutdown), and normal shutdown. There shall be no electrical interferences caused by the transmission or controller beyond the limits referenced in MIL-STD-826.

4.3.2.1.15 Drive disconnect. The drive coupling shall be disengaged a total of 20 times at maximum rated input speed and 0 load and 5 times at minimum rated input speed and maximum rated load. The disconnect device should be tested with maximum rated torque on the input shaft. This occurs at maximum load and minimum speed. Provisions shall be incorporated in the drive assembly to reengage the drive coupling after the assembly is stopped. There shall be no excessive wear on the drive coupling mating parts as a result of this test.

4.3.2.1.16 Shear section. Two samples of the constant speed drive shear section, stabilized at room temperature, shall be torsion tested to determine that the torque at which shearing occurs conforms to the requirement specified in the model specification.

4.4 Quality conformance tests. Each drive assembly submitted for delivery under a procurement contract shall be subjected to the quality conformance examinations and tests specified herein. Acceptance or approval of material during manufacture shall, in no case, be construed as guarantee of the acceptance of the finished product.

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a. Examination of product (4.3.2.1.2)

b. No load (4.3.2.1.3)

c. Rated load (4.3.2.1.4)

d. Speed regulation (4.3.2.1.5)

e. Parallel operation (4.3.2.1.6)*

f. Load division (4.3.2.1.7)*

g. Underspeed switch (4.3.2.1.9)*

*To be accomplished on first twenty drives and two out of 100 drives produced thereafter.

4.4.1 Drive disconnect short test. The drive coupling shall be disengaged while the assembly is being driven at rated output speed and 0 load. The coupling should be reengaged after the assembly is stopped. The rated load test 4.3.2.1.4 should be repeated.

4.5 Reliability tests

4.5.1 Inherent reliability assurance tests

4.5.1.1 Purpose of tests. These tests shall provide a minimum and practical basis for:

a. The disclosure of design and quality control deficiencies under conditions advantageous for analysis, and providing out corrective action.

b. Comparing the relative inherent reliability of different model drive assemblies.

c. Establishing a quantitative measurement of inherent reliability. In order to achieve the maximum degree of assurance within reasonable limits of time and costs expended, the frequency of high level load cycles has been intensified and the number of conditions imposed have been minimized. The tests outlined are intended to represent the more critical of the design conditions without exceeding the levels of drive load, speed, and temperature likely to be encountered in service.

4.5.1.2 Demonstration of reliability. The required reliability (MTBF) shall be demonstrated at least once and periodically as specified in the model specification. MIL-STD-781 shall be followed. If the discrimination ratio =3.0, its Test Plan XXIII shall be used; Test Plan XXIV may be used if it is specified in the model specification submitted before the test. A minimum of three units shall be tested. One unit shall be designated before the test for testing to a minimum of 1000 hours.

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If representative production units are not available as needed, some of the test units may be pre-production units provided they are sufficiently representative of production units. All production units shall together account for at least 300 reliability test hours. Units shall not be overhauled during the test more frequently than those for normal service. All failures, including excursions of test data beyond established limits shall be immediately reported to the procuring activity. Unless otherwise qualified by the model specification, each adjustment or removal of a test unit or part from test in the face of impending drift out of specification limits or impending failure shall be reported and charged one test failure. New, overhauled, or repaired units shall not be subjected to substantially more testing before reliability testing than that given to every production unit. Additional reliability test conditions shall be as specified for the endurance tests (4.3.2.1.12). The endurance and reliability tests may be combined if this is specified in the model specification submitted before the tests are begun. Units subjected to reliability tests may be overhauled and delivered unless otherwise specified in the model specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Unless otherwise specified by the contract or purchase order, or by the detail specification, each constant speed drive shall be flushed with the fluid specified in the model specification. All parts shall be sealed with closures conforming to MIL-C-5501. The component shall be wrapped or bagged with grade A, greaseproof material conforming to MIL-F-22191, and secured with tape conforming to PPP-T-60. Each wrapped component shall be cushioned and secured with special die cut dunnage, molded dunnage, or mounting devices within a container conforming to either PPP-B-601, PPP-B-636, MIL-C-6054, MIL-C-9282, or MIL-B-25305. The component shall be protected from corrosion in accordance with method II of MIL-P-116. The choice of container and submethod of preservation to be consistent with the weight, cube, and characteristics of the component. The unit container shall also serve as the shipping container.

5.1.2 Level C. The component shall be packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to first receiving activity for immediate use. This level may conform to the manufacturers commercial practice, provided the latter meets the requirements of this level.

5.2 Packing

5.2.1 Level A and Level B. The components preserved and packaged as specified in 5.1.1 shall meet the level of packing as specified by the procuring activity.

5.2.2 Level C. Components, packaged as specified in 5.1.2, shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

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5.3 Marking of shipment. In addition to any special marking required by the contract or order, or the detail specification, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Reinspection marking. Reinspection marking shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The constant speed drives covered by this specification are intended for use in driving electrical generators in aircraft installations.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification.

b. Type of system in which the constant speed drive is to be used (see 1.1).

c. Manufacturer's part No.

d. Selection of applicable levels of packaging and packing (see section 5).

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 1650-F016

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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